Answer key: Lesson 5, Make-a-Ten Adding

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Fill the first ten-frame by moving counters from the second one. Then add.

Answer: 1 more; 10 and 3; 9 + 4 = 13

The first frame has 1 empty space, so 9 needs 1 more. Move 1 counter from the 4. The first frame is full (10) and 3 are left. 10 + 3 = 13.

Watch for: Writing 10 and 4: forgetting to take the moved counter away from the 4.; Counting all the counters again and slipping (12 or 14).

S2. Draw 7 dots and 6 dots in the box. Ring ten dots to make a ten. Then write the sums.

Answer: 7 + 6 = 10 + 3 = 13 (ten dots ringed, 3 left over)

Ring the 7 dots and 3 of the 6 dots to make ten. The other 3 dots are outside the ring. So 7 + 6 = 10 + 3 = 13.

Watch for: Ringing ten dots but forgetting the dots left outside.; Miscounting when counting every dot one by one.

S3. How many more make the next ten?

Answer: 3, 2, 4, 1

(a) 7 and 3 make 10. (b) 18 is 1 ten and 8 ones, and 8 and 2 make 10, so 18 + 2 = 20. (c) 6 and 4 make 10, so 26 + 4 = 30. (d) 9 and 1 make 10, so 39 + 1 = 40.

Watch for: Writing the next ten itself (20 for part b).; Counting from the tens digit instead of the ones digit.

S4. Add 6 to 28. Make a ten first.

Answer: 2 more; 6 = 2 and 4; 28 + 6 = 34

28 needs 2 more to make 30. Split 6 into 2 and 4. 28 + 2 = 30, and 30 + 4 = 34.

Watch for: Writing 36: adding all 6 after reaching 30, when 2 of the 6 were already used.; Splitting 6 into 2 and 3 or another wrong pair.

S5. Add 7 to 35. Make a ten first.

Answer: 5 more; 35 + 7 = 42

35 needs 5 more to make 40. 7 is 5 and 2. 35 + 5 = 40, and 40 + 2 = 42.

Watch for: Writing 47 (adding 7 to the 4 tens).; Forgetting the 2 left over after the ten.

S6. Add. Make a ten first if it helps.

Answer: 15, 23, 53, 64

(a) 9 + 1 = 10, 10 + 5 = 15. (b) 18 + 2 = 20, 20 + 3 = 23. (c) 47 + 3 = 50, 50 + 3 = 53. (d) 56 + 4 = 60, 60 + 4 = 64.

Watch for: Moving the right amount to make the ten but then adding the whole second number.; Making a ten with the wrong number (for example 9 + 6 = 9 + 4 + 2).

S7. Siti adds 8 + 6. She writes: 8 + 2 = 10, then 10 + 6 = 16. Is Siti right? Write the right answer. Say what she forgot.

Answer: No. 14. She moved 2 from the 6 to make the ten, so only 4 are left: 10 + 4 = 14.

8 + 2 = 10 uses 2 of the 6. Only 4 are left, so the sum is 10 + 4 = 14. Siti added all 6 again.

Watch for: Saying she is right because 10 + 6 looks like a good sum.; Writing 14 without saying what she forgot (a spoken reason is fine).

Think harder (page 3)

T1. Mei adds 4 + 9 + 6. She finds two numbers that make 10 first. Which two? What is the total?

Answer: 4 and 6 make 10. The total is 19.

4 and 6 make 10. 10 + 9 = 19. Check the long way: 4 + 9 = 13, and 13 + 6 = 19.

Watch for: Adding the first two numbers (4 + 9) and stopping or slipping.; Thinking the pair must sit side by side.

T2. Add 8 + 5 + 2 + 7. Find the pairs that make a ten. Write your pairs.

Answer: 22 (8 + 2 = 10, 5 + 7 = 12, 10 + 12 = 22)

8 + 2 = 10. 5 + 7 = 12 (5 + 5 + 2). 10 + 12 = 22.

Watch for: Adding from left to right and losing track.; Using a number twice or leaving one out.

T3. Three sums make 30. One does not. Which one? 17 + 13, 18 + 12, 19 + 12, 16 + 14. Say how you know without adding every one.

Answer: 19 + 12 (it makes 31). The ones 9 and 2 make 11, not 10.

7 + 3 = 10, 8 + 2 = 10 and 6 + 4 = 10, so those make 30. 9 + 2 = 11, so 19 + 12 is 31.

Watch for: Adding every sum one by one and slipping.; Choosing 18 + 12 because 18 looks big.

T4. Ben has 18 red marbles, 34 blue marbles and 12 green marbles. How many marbles has he in all? Show your pair.

Answer: 64 (18 + 12 = 30, 30 + 34 = 64)

18 + 12 = 30. 30 + 34 = 64. The order does not change the total.

Watch for: Leaving out one group of marbles.; Adding 18 + 34 first and slipping on the ten.

Go further (page 4)

G1. Add 18 + 7 + 12 + 23 + 3 + 17. Look for pairs that make a ten or a tens number. Show your pairs.

Answer: 80 (18 + 12 = 30, 7 + 3 = 10, 23 + 17 = 40)

18 + 12 = 30. 7 + 3 = 10. 23 + 17 = 40. 30 + 10 + 40 = 80. Another way: 7 + 23 = 30, 18 + 12 = 30, 3 + 17 = 20, total 80.

Watch for: Adding left to right and losing the running total.; Using a number twice or leaving one out.

G2. Add the numbers 1 to 9: 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9. Look for pairs that make 10. Then find 1 to 10.

Answer: 45 and 55 (four pairs of 10, and the 5 is left)

1 + 9, 2 + 8, 3 + 7 and 4 + 6 make four tens: 40. The 5 has no partner. 40 + 5 = 45. Adding 10 makes 55.

Watch for: Forgetting the 5 in the middle has no partner.; Making only three pairs.

G3. Write a sum of 6 numbers that makes 100. Use pairs that make a ten or a tens number. Show your pairs.

Answer: Any six numbers that total 100. For example 45 + 5 + 20 + 17 + 3 + 10 = 100 (45 + 5 = 50, 17 + 3 = 20, 20 + 10 = 30).

Choose pairs first: 50 + 20 + 30 = 100. Then split each into two numbers, for example 45 + 5, 17 + 3, 20 + 10. Add them to check.

Watch for: Choosing numbers that total 90 or 110: add every number to check.; Using fewer than 6 numbers.

Investigation (page 5): Move the fewest

Fill up the bigger number: it has the fewest empty spaces, so you move the fewest dots. The number moved is 10 minus the bigger number. 9 + 6: move 1, not 4. 8 + 7: move 2, not 3. If the two numbers are the same (7 + 7), either way moves 3.

Look for: A child who tries several pairs, records how many dots moved each way, and notices the bigger number is closer to ten. Credit any clear reason, such as the bigger number has fewer empty spaces.

Exit ticket (page 5)

E1. Add by making a ten.

Answer: 13

7 + 3 = 10. 6 is 3 and 3, so 10 + 3 = 13.

Watch for: Writing 16 (10 + 6).; Writing 12 or 14 after counting on fingers.

E2. Add by making a ten.

Answer: 43

38 + 2 = 40. 5 is 2 and 3, so 40 + 3 = 43.

Watch for: Writing 45 (40 + 5).; Writing 33 or 83.

E3. Add 6 + 9 + 4. Which two numbers did you add first?

Answer: 19. 6 and 4 make 10, then 10 + 9 = 19.

6 + 4 = 10. 10 + 9 = 19.

Watch for: Adding 6 + 9 first (15) and then slipping.; Not noticing that 6 and 4 make a ten.

Sneak-peek puzzles (not in the workbook)

1. 12 stars.
2. Yes. Nothing was added or taken away. 9 + 5 = 14 and 10 + 4 = 14.